



# Kubernetes Notes



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# Kubernetes Components

**Pod :** The smallest deployable unit in Kubernetes, representing a group of one or more containers that share network and storage resources. Pods are the atomic unit of scheduling and run on a single node.

**Node :** A worker machine in the Kubernetes cluster that hosts pods. Nodes are the fundamental computing hardware units that provide computational resources for running containerized applications.

**Objects :** Persistent entities in the Kubernetes system that represent the state of the cluster, including deployments, services, pods, and other resources that define the desired cluster configuration.

**Monitoring :** Tracks resource utilization and performance of nodes, pods, and containers using commands like `kubectl top` to provide insights into cluster health and resource consumption.

**Deployment :** A Kubernetes object that manages the lifecycle of pods, ensuring a specified number of replica pods are running and facilitating updates and rollbacks of application versions.

**Service :** An abstraction that defines a logical set of pods and a policy to access them, providing network connectivity and load balancing for applications across the cluster.

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**Ingress** : A Kubernetes object that manages external access to services, typically HTTP/HTTPS, providing routing, SSL termination, and name-based virtual hosting capabilities.

**Endpoints** : Represents the network addresses of pods backing a service, dynamically updated to track which pods are available to handle requests.

**DaemonSet** : Ensures a specific pod runs on all (or selected) nodes in the cluster, typically used for cluster-wide services like monitoring, logging, or node-level system daemons.

**Jobs** : Kubernetes objects that create one or more pods to perform a specific task and ensure the task completes successfully, useful for batch processing and one-time computational tasks.

**Rollout** : Manages the deployment and update process, allowing controlled updates, rollbacks, and tracking of deployment revisions.

**Secret** : Kubernetes objects used to store and manage sensitive information like passwords, OAuth tokens, and SSH keys, providing a secure way to handle confidential data in the cluster.



# Pod Commands

## Basic Pod Commands

- `kubectl get pod`: List all pods in current namespace
- `kubectl get pod -o wide`: Detailed pod information with node details
- `kubectl get pod -w`: Watch live pod status updates
- `kubectl get pod -o yaml`: Export pod configuration in YAML format
- `kubectl describe pod <pod_name>`: Detailed pod status and events
- `kubectl delete pod <pod_name>`: Remove specific pod

## Advanced Pod Commands

- `kubectl edit pod <pod_name>`: Modify pod configuration live
- `kubectl logs <pod_name>`: View pod logs
- `kubectl exec -it <pod_name> /bin/bash`: Interactive shell inside pod

## Additional Useful Commands

- `kubectl get pods --show-labels`: List pods with their labels
- `kubectl get pods --field-selector=status.phase=Running`: Filter running pods
- `kubectl top pod`: Resource usage metrics
- `kubectl port-forward <pod_name> <local_port>:<pod_port>`: Network port forwarding



# Node Commands

## Node Management Commands

- `kubectl describe node <node_name>`: Detailed node information and status
- `kubectl get node <node_name>`: Get basic node details
- `kubectl get node <node_name> -o yaml`: Export node configuration in YAML format
- `kubectl drain node <node_name>`: Prepare node for maintenance by evicting all pods
- `kubectl cordon node <node_name>`: Mark node as unschedulable for new pods
- `kubectl uncordon node <node_name>`: Mark node as schedulable again

## Additional Useful Commands

- `kubectl get nodes`: List all cluster nodes
- `kubectl top node`: View node resource usage
- `kubectl label node <node_name>`: Add or modify node labels

# Object Creation Commands

## Resource Creation

- `kubectl apply -f <file_name>.yaml`: Create resource from YAML file
- `kubectl apply -f <file1>.yaml -f <file2>.yaml`: Create from multiple files
- `kubectl apply -f ./<directory_name>`: Create all files in directory
- `kubectl apply -f https://<url>`: Create from URL

## Pod Creation

- `kubectl run <pod_name> --image=<image_name>`: Create basic pod
- `kubectl run <pod_name> --image=<image_name> --port <port> --expose`: Create pod and expose as service
- `kubectl run <pod_name> --image=<image_name> --dry-run=client -o yaml > <file_name>.yaml`: Generate Pod YAML file

## Deployment Creation

- `kubectl create deployment <name> --image=<image_name>`: Create deployment
- `kubectl create deployment <name> --image=<image_name> --dry-run=client -o yaml > <file_name>.yaml`: Generate Deployment YAML file

## ConfigMap Creation

- `kubectl create configmap <name> --from-literal=<key>=<value>`: Create from key-value pairs
- `kubectl create configmap <name> --from-file=<file_name>`: Create from file
- `kubectl create configmap <name> --from-env-file=<file_name>`: Create from environment file

## Secret Creation

- `kubectl create secret generic <name> --from-literal=<key>=<value>`: Create from key-value pairs
- `kubectl create secret generic <name> --from-file=<file_name>`: Create from file

# Monitoring Usage Commands

## Node Resource Monitoring

- `kubectl top node`: Display resource usage for all nodes
- `kubectl top node <node_name>`: Get specific node's CPU and memory utilization

## Pod Resource Monitoring

- `kubectl top pod`: Display resource usage for all pods
- `kubectl top pod <pod_name>`: Get specific pod's CPU and memory utilization

## Additional Monitoring Commands

- `kubectl top pod -n <namespace>`: View pod resources in specific namespace
- `kubectl top pod --sort-by=cpu`: Sort pods by CPU usage
- `kubectl top pod --sort-by=memory`: Sort pods by memory usage

# Deployment Commands

## Basic Deployment Commands

- `kubectl get deployment`: List all deployments
- `kubectl get deployment <deployment_name>`: Get specific deployment
- `kubectl get deployment <deployment_name> -o yaml`: Get deployment in YAML format
- `kubectl get deployment <deployment_name> -o wide`: Detailed deployment information

## Deployment Management

- `kubectl edit deployment <deployment_name>`: Modify deployment configuration
- `kubectl describe deployment <deployment_name>`: Detailed deployment status
- `kubectl delete deployment <deployment_name>`: Remove deployment
- `kubectl scale deployment <deployment_name> --replicas=<number>`: Change deployment replica count

# Service Commands

## Service Management Commands

- `kubectl get service`: List all services in current namespace
- `kubectl get service <service_name>`: Get specific service details
- `kubectl get service <service_name> -o yaml`: Get service in YAML format
- `kubectl get service <service_name> -o wide`: Detailed service information
- `kubectl edit service <service_name>`: Modify service configuration
- `kubectl describe service <service_name>`: Detailed service status
- `kubectl delete service <service_name>`: Remove service from cluster



# Ingress Commands

## Ingress Management Commands

- `kubectl get ingress`: List all ingresses in current namespace
- `kubectl get ingress -o yaml`: Get ingress in YAML format
- `kubectl get ingress -o wide`: Detailed ingress information
- `kubectl edit ingress <ingress_name>`: Modify ingress configuration
- `kubectl describe ingress <ingress_name>`: Detailed ingress status
- `kubectl delete ingress <ingress_name>`: Remove ingress from cluster



# Endpoints Commands

## Endpoints Management Commands

- `kubectl get endpoints`: List all endpoints in current namespace
- `kubectl get endpoints <endpoints_name>`: Get specific endpoint details
- `kubectl get endpoints -A`: List endpoints across all namespaces



# DaemonSet Commands

## DaemonSet Management Commands

- `kubectl get daemonset`: List all daemonsets in current namespace
- `kubectl get daemonset <daemonset_name>`: Get specific daemonset details
- `kubectl get daemonset <daemonset_name> -o yaml`: Get daemonset in YAML format
- `kubectl edit daemonset <daemonset_name>`: Modify daemonset configuration
- `kubectl describe daemonset <daemonset_name>`: Detailed daemonset status
- `kubectl delete daemonset <daemonset_name>`: Remove daemonset from cluster



# Job Commands

## Job Management Commands

- `kubectl get job`: List all jobs in current namespace
- `kubectl get job <job_name>`: Get specific job details
- `kubectl get job <job_name> -o yaml`: Get job in YAML format
- `kubectl edit job <job_name>`: Modify job configuration
- `kubectl describe job <job_name>`: Detailed job status
- `kubectl delete job <job_name>`: Remove job from cluster



# Rollout Commands

## Deployment Rollout Management

- `kubectl rollout restart deployment <deployment_name>`: Restart deployment
- `kubectl rollout undo deployment <deployment_name>`: Undo to previous deployment revision
- `kubectl rollout undo deployment <deployment_name> --to-revision=<revision_number>`: Undo to specific deployment revision
- `kubectl rollout history deployment <deployment_name>`: View all deployment revisions
- `kubectl rollout history deployment <deployment_name> --revision=<revision_number>`: Get details of specific revision



# Secret Commands

## Secret Management Commands

- `kubectl get secret`: List all secrets in current namespace
- `kubectl get secret <secret_name>`: Get specific secret details
- `kubectl get secret <secret_name> -o yaml`: Get secret in YAML format
- `kubectl describe secret <secret_name>`: Detailed secret information
- `kubectl delete secret <secret_name>`: Remove secret from cluster
- `kubectl edit secret <secret_name>`: Modify secret configuration